TIER I ADOPTION NOTICE

Robotics & Autonomous Systems, Ground (RAS-G) Interoperability Profile (IOP) Version 6.0 (RAS-G-IOP-V6.0) was adopted on 01 October 2024 for use by the Department of Defense (DoD). Proposed changes by DoD activities must be submitted to the DoD Adopting Activity: Program Executive Office (PEO) Combat Support & Combat Service Support (CS&CSS). Information regarding this technical standard, including copies of the latest revision, may be obtained electronically from either of the following sites, depending on the user.

- DoD Common Access Card (CAC) Holders: https://armyeitaas.sharepoint-mil.us/sites/usaasc-peocscss-fp-ENG/Systems%20Engineering/Forms/AllItems.aspx?csf=1&web=1&e=VMIeoQ&CID=5c44beb7%2Dad76%2D41fc%2D9006%2Da7d3fa0b3942&FolderCTID=0x012000F11C7F0827E72449A0666A33A8F76928&id=%2Fsites%2Fusaasc%2Dpeocscss%2Dfp%2DENG%2FSystems%20Engineering%2FRobotics%2FRAS%2DG%20IOP&viewid=9182fae4%2D1c67%2D49c7%2D93e7%2D303f5488c793
- Industry Members of the National Advanced Mobility Consortium (NAMC): https://namcgroups.org/ras-coi-member-page/

This standard implements STANREC 4818, Robotic Autonomous Systems – Ground Interoperability Profile – Allied Engineering Publication (AEP)-4818. When changes to or cancellation of this standard is proposed, the Adopting Activity should coordinate the action with the U.S. National Point of Contact for the international standardization agreements identified in the ASSIST Online database at https://assist.dla.mil.

The DoD launched an initiative to identify and define interoperability standards to be organized and maintained within a Robotics and Autonomous Systems – Ground (RAS-G) Interoperability Profile (IOP). This IOP will be employed by Unmanned Ground Vehicle (UGV) acquisition managers in the acquisition of future Programs of Record, the upgrade of fielded systems, and the evaluation/acquisition of Commercial-Off-The-Shelf (COTS) products. Throughout this document, the terms RAS-G and UGV are used interchangeably.

RAS-G-IOP V6.0 is published by the Army through partnership with industry partners via the National Advanced Mobility Consortium (NAMC). NAMC brings together Government, Industry, and Academic organizations to advance ground mobility technologies, drive relevant and rapid solutions, and empower the U.S. warfighter. The IOP defines the interoperable interfaces and protocols necessary to enable interoperability and modularity. The exact set of capabilities addressed in IOP V6 is described in the RAS-G IOP V6 Capabilities Plan. Periodic revisions to the IOP will be performed in order to expand and evolve its scope as necessary, based on the evolution of Warfighter capability requirements and technological advances.

A primary goal of this initiative is to leverage existing and emerging standards within the Unmanned Vehicle (UxV) community such as the Society of Automotive Engineers (SAE) AS-4 Joint Architecture for Unmanned Systems (JAUS) standard, and the Controller Area Network (CAN) SAE J1939 standard. The Navy Advanced Explosive Ordnance Disposal Robotic System (AEODRS) Architecture Description Documents (which use SAE AS-4) has been leveraged significantly. The end goals of this effort are:

- Facilitating interoperability among new UGV initiatives and legacy systems
- Facilitating interoperability between controllers and UxV robotic system(s)
- Facilitating collaboration between UGV and UAS systems

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- Providing a path forward to standardized interoperable technology solutions
- Promoting payload and on-board subsystem modularity and commonality across the portfolio of UGV systems

The majority of IOP documents are approved for Distribution A. There are some additional annexes that are controlled as Controlled Unclassified Information (CUI). Additionally, the IOP provides a data dictionary of Controller Area Network (CAN) messages in the SAE J1939 format for communicating between autonomous mobility kits and heavy trucks (native as a .DBC file). This can be provided upon request to authorized organizations by the Army DEVCOM Ground Vehicle Systems Center (GVSC). There is also an IOP Conformance Verification Tool (CVT) available from GVSC. The IOP documentation structure is as follows:

- **RAS-G IOP – Overarching Profile:** In addition to the topics specified above, this document also provides separately published attachments that comprise the Overarching IOP V6. These attachments are described below.
- **RAS-G IOP V6 – Capabilities Plan:** Defines capability requirements related to the employment and usage of UGVs to perform current and relevant near-term robotic missions, in turn scoping and bounding the content of the RAS-G IOP. This Capabilities Plan is based on a mission analysis that reviewed the usage of currently fielded and near term UGVs, and the capability requirements of a number of existing and emerging Programs of Record.
- **RAS-G IOP – SAE JAUS Profiling Rules:** Specifies the manner in which the SAE AS-4 JAUS standards have been profiled, to include clarification or additional content to define interoperability between controllers and UGVs as well as intra-UGV (platform/subsystem) interoperability.
- **RAS-G IOP – Custom Services, Messages and Transports:** Specifies additional, custom SAE AS-4 JAUS messages and transport protocols required to support the scope of the RAS-G IOP. Although titled "custom", these messages are published and standardized within the RAS-G IOP community with the end goal of transitioning to the SAE AS-4 JAUS standard(s) or other standards bodies for official adoption.
- **RAS-G IOP – Control Profile:** This document specifies the Operator Control Unit (OCU) logical architecture, standards, Human-Machine Interface (HMI) requirements, and conformance approach to include host application user interface requirements, such as mission planning and command and control. Although OCU concepts and high-level architecture are touched upon in the Overarching Profile, the Control Profile provides the more detailed requirements to specify how interoperability is to be achieved for conformant controllers.
- **RAS-G IOP – Payloads Profile:** This document specifies the payload classification, standards, requirements, and conformance approach. Although these concepts are touched upon in the Overarching Profile, the Payloads Profile provides the more detailed requirements to specify the interoperability requirements for payloads with respect to the UGV platform.
- **RAS-G IOP – Communications Profile:** This document specifies the communications standards, requirements, and conformance approach. Although these concepts are touched upon in the Overarching Profile, the Communications Profile provides the more detailed requirements to specify interoperability requirements for communications between and among controllers and UGVs.
- **RAS-G IOP - Applique Profile:** This document specifies the appliqué systems classification, standards, requirements, and conformance approach. Although these concepts are touched upon in the Overarching Profile, the Applique Profile provides the more detailed requirements to specify

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the interoperability requirements for appliqué systems with respect to the unmanned ground systems, controllers, and base manned vehicle systems.

- **RAS-G IOP – Engagement Profile:** This document (new as of V5) provides specific rules, requirements, guidance, and considerations associated with the profiling of contact engagement systems (i.e., weapons systems, non-lethal, object tracking, etc) on robots and vehicles to achieve system-level and platform-level interoperability.

Adopting Activity:
Army – CSCS
(Project MOSS-24-012)

Reviewing Activities:
Army – AC, AR, AT, AV, CR6, MI
Navy –EC, MC, NM
Air Force – 01,20,
OSD - SE

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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